Lesson 12: Hearing It Happen

Unreal Engine (optional advanced track)

VIDEO GAME CREATION

This week: your game finally sounds like a game. Four moments - firing, hitting, clearing a wave, and losing - each get their own distinct cue.

Open your own [YourName]Game project from Lesson 11 - don't start a new one.

If you already built an optional sound idea in an earlier lesson (Lessons 2, 3, 4, or 6), keep it - today reuses it directly instead of replacing it. If you didn't, today's steps are written as if starting fresh.

Setup

1. Open your [YourName]Game project from Lesson 11.
2. Import sfx_wave.wav as a Sound Wave asset - today's one genuinely new sound - alongside your existing sfx_shoot.wav, sfx_hit.wav, sfx_lifelost.wav, sfx_gameover.wav, and (if built) sfx_win.wav.

Level 1 - Minimal (MVP)

Four sounds fire at the correct moment, at least once each: a shot, a hit/explosion, a wave-clear (on any non-final wave), and a game over. No silence, no obviously wrong triggers.

3. Confirm your IA_Fire Enhanced Input Action Event already has a Play Sound 2D(sfx_shoot.wav) node chained after Spawn Actor from Class (Lesson 3) - add it now if it's missing.
4. In your bullet-enemy collision handling, confirm Play Sound 2D(sfx_hit.wav) already fires after the successful Cast (Lesson 4) - add it now if it's missing, before the win-check.
5. In HandleEnemyDestroyed's nested Branch (from Lesson 11), add Play Sound 2D(sfx_win.wav) in the True case (only if you built it) and Play Sound 2D(sfx_wave.wav) in the False case.
6. In HandleEnemyEscaped, confirm Play Sound 2D(sfx_gameover.wav) already fires inside the Lives <= 0 Branch (from Lesson 8) - add it now if it's missing. This sound is mandatory starting today even if you skipped it before.
7. Play in Editor and play through a full round - shoot, hit, an early wave clear, and a loss should each sound distinct.

Highlighted lines are new this week; everything else is code you already have.

```
Custom Event HandleEnemyDestroyed (revised) ->
  EnemiesRemaining = EnemiesRemaining - 1
  Score = Score + 10
```

```
  Play Sound 2D(sfx_hit.wav)
  Call UpdateHUD(HUDWidget, Score, Lives, Wave)
  Branch(EnemiesRemaining <= 0) -> True:
    Branch(Wave >= TotalWaves) -> True:
      GameActive = false
      Play Sound 2D(sfx_win.wav) // your Lesson 6 Level 3 sound
      Call SetStatus(HUDWidget, "You Win! Press R to Restart")
    False:
      Wave = Wave + 1
      EnemiesRemaining = 5
      Play Sound 2D(sfx_wave.wav)
      Call UpdateHUD(HUDWidget, Score, Lives, Wave)
      Call SpawnWave

Custom Event HandleEnemyEscaped (unchanged from Lesson 8) ->
  Lives = Lives - 1
  Branch(Lives <= 0) -> True:
    GameActive = false
    Play Sound 2D(sfx_gameover.wav) // your Lesson 8 Level 3 sound,
mandatory as of today
    Call SetStatus(HUDWidget, "Game Over - Press R to Restart")
  False:
    Play Sound 2D(sfx_lifelost.wav) // your Lesson 8 Level 3 sound
```

Alternate Path: C++ (code, optional advanced track)

The same sound calls work in C++ instead of Blueprint nodes.

8. Add UGameplayStatics::PlaySound2D(GetWorld(), HitSound) at the same spot as the Blueprint hit sound, and matching calls for WinSound/WaveSound inside your kill-handling function's nested if/else.
9. Confirm UGameplayStatics::PlaySound2D(GetWorld(), GameOverSound) already fires inside your escape-handling function's lives<=0 branch (from Lesson 8), with LifeLostSound in the else case - add them now if either is missing.

This mirrors the Blueprint path's Play Sound 2D nodes exactly - same moments, just called from code.

Level 2 - Complete (Alpha/Beta)

All four sounds fire reliably every time, with no missing or duplicate playback, and each is clearly distinguishable from the others by ear.

- Play a full round: shoot, hit, an early wave clear, and a loss each produce a distinct sound.
- If you built the optional win sound, confirm it still plays on the true win, and sfx_wave.wav doesn't also fire on that same frame.
- Play Sound 2D nodes layer naturally - confirm rapid shooting and a hit landing together both play correctly.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Audio: if you haven't already, add the optional win sound now - it slots into the exact same Play Sound 2D pattern as today's wave sound, just a different Branch case.

- Audio: use a Random Float in Range node to vary the Pitch Multiplier parameter on the hit sound's Play Sound 2D node for slight variety.
- Structure: notice every Play Sound 2D node this week lives directly inside the exact Custom Event representing its trigger - no separate audio-manager Blueprint needed for a game this size.

If Something's Wrong

- No sound on wave-clear: confirm the sfx_wave.wav Play Sound 2D node is actually wired inside the False case of the nested Branch, not before it.
- Game-over sound plays repeatedly: confirm the Branch is still a one-time check tied to GameActive, matching Lesson 8's pattern.
- No sound at all: confirm the imported asset is actually a Sound Wave (not left as a raw .wav in the Content Browser needing re-import).

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